

Forward, Futures, FX Spot and Forwards

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Probability Measures in Finance

1. Real-World Measure (P):

1. **Intuition:** Reflects the actual probabilities of events occurring in the real world.
2. **Use:** Used for historical analysis and risk management.
3. **Example:** The probability of rain tomorrow based on weather patterns.

2. Risk-Neutral Measure (Q):

1. **Intuition:** Assumes investors are indifferent to risk, allowing for the pricing of derivatives by discounting expected payoffs at the risk-free rate.
2. **Use:** Used for pricing financial derivatives.
3. **Example:** Calculating the fair price of a stock option by assuming the expected return is the risk-free rate.

3. Forward Measure (T):

1. **Intuition:** A measure where the numeraire is a zero-coupon bond maturing at a specific future time T , simplifying the pricing of derivatives with payoffs at T .
2. **Use:** Used for pricing interest rate derivatives and other instruments with specific maturity payoffs.
3. **Example:** Pricing a bond option with a payoff at maturity.

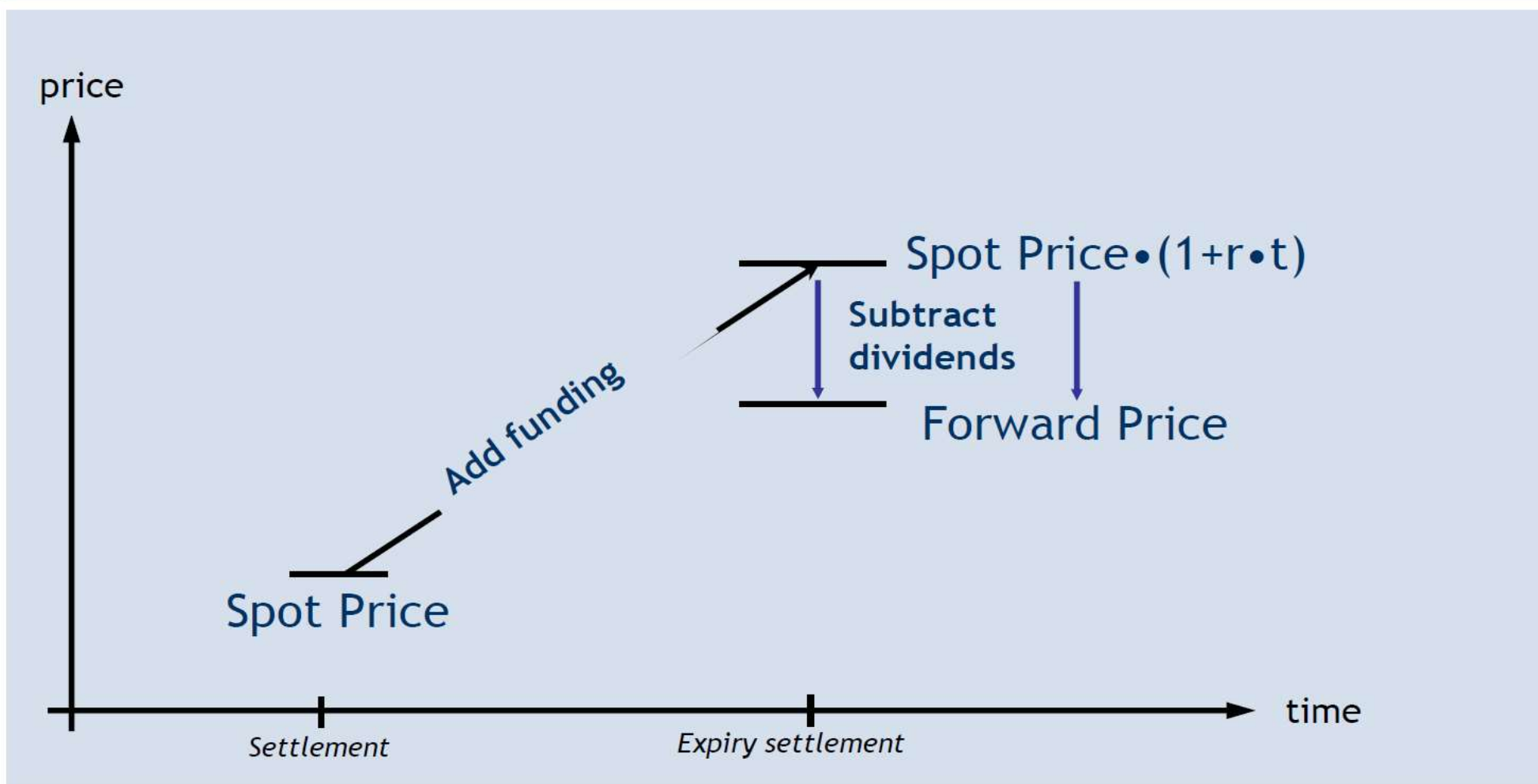
Radon-Nikodym Derivative

Mathematical tool used to change from one probability measure to another by adjusting for the change in the numeraire.

Example: Suppose you have a probability measure for a fair coin (PP) and want to change it to a biased coin measure (QQ) where heads are twice as likely. The Radon-Nikodym derivative would adjust the probabilities accordingly, doubling the weight of heads.

Futures / forward pricing

The fair price to deliver a stock or index at a future date should be equal to the cost of carry minus any dividends paid out (i.e. the cost to deliver the stock at expiry).



Forward Stock Price

$$[\text{Theoretical Futures Price}] = [\text{Spot}] \times [1 + R \times T] - D$$

$$[\text{Forward Price}] = \text{Spot} + [\text{Spot} \times R \times T] - D$$

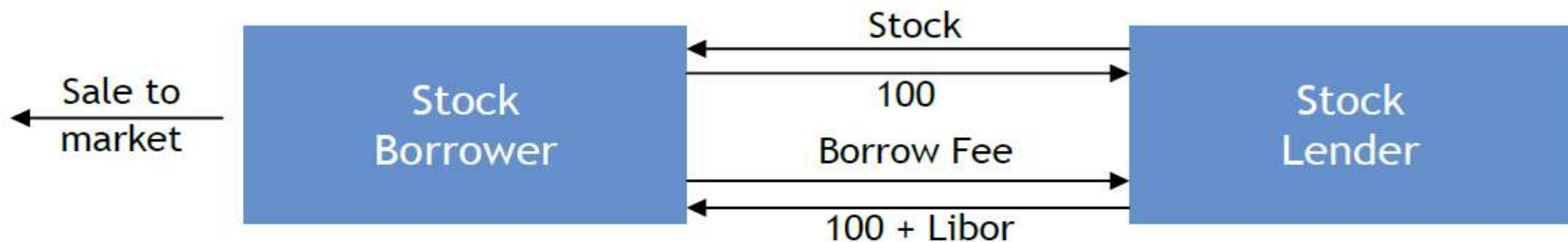
where:

Spot	=	Spot price of underlying
R (financing rate)	=	Short term borrowing rate used to finance stock positions minus any stock borrow
T	=	Time measured in years: days from settlement to expiry settlement / 360 (365)
D	=	Future value of dividends from settlement to expiration

Buy futures	=	buy at forward price
Buy and lend stock	=	buy at forward price
Receive on stock swap	=	buy at forward price

Stock Borrow

Stylised Stock Borrow



Today

Buy stock

Cashflow

-100

Stock flow

+ stock

Lend stock

+100

- stock

End of Borrow

Receive stock

$-100 * [1 + (\text{Libor} - \text{Borrow}) \times T]$

+stock

Buy and lend stock

= forward purchase at forward price

Sell and borrow stock

= forward sale at forward price

Example: Calculating a stock forward

Royal Dutch 2-month forward price: 16-Apr-04

Royal Dutch stock is trading at €41.65 with a borrow of 0.5%

Buy and lend stock for 3 months

Cash dividend of €1.02, XD 26 Apr 2004

Euribor @ 2.12%

$$\begin{aligned}\Rightarrow \text{Forward price} &= 41.65 * (1 + (0.0212 - 0.0050) * 63/360) - 1.02 \\ &= 41.65 + 0.12 - 1.02 \\ &= 40.74\end{aligned}$$

$$\text{Futures price} = 40.68$$

(Indicative levels as of 16 Apr 2004)

Vanilla Equity Futures vs. Forwards

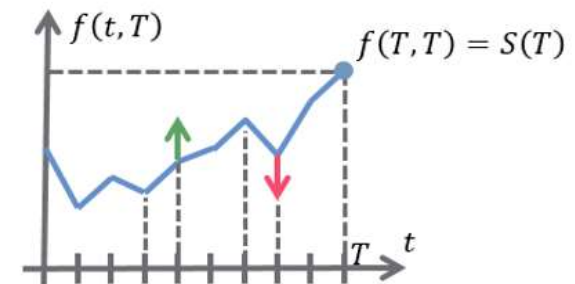
It's a common misconception to think of futures and forwards as being the same. There are indeed fundamental differences:

- A **Futures Contract** is an agreement between two parties to buy or sell a certain underlying at a fixed time T in the future *where payments are made “continuously” in time via a variation margin approach.*
- A **Forward Contract** is an agreement between two parties to buy or sell a certain underlying at a fixed time T in the future *at a (strike) price K agreed upon at inception of the contract.*

The *Futures Price* $f(t, T)$ is the quoted mark-to-market of the contract. During an arbitrary time interval $(s, t]$, the holder of the contract receives the amount $f(t, T) - f(s, T)$.

The futures price is computed as the expectation, under the risk neutral measure of the underlying price at maturity.

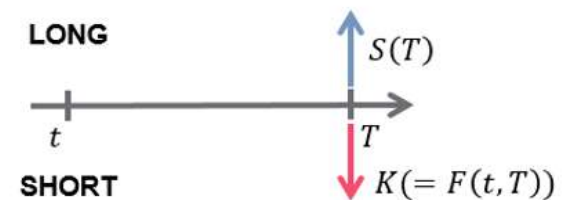
$$f(t, T) = E_t^{\mathbb{Q}}[S(T)]$$



The *Forward Price* $F(t, T)$ is the amount of money, determined at the inception of the contract t and that will be exchanged at expiry T , such that the value of the forward contract equals zero at time t . In other words, it is the “fair strike”.

$$F(t, T) = \frac{1}{D(t, T)} E_t^{\mathbb{Q}} \left[e^{-\int_t^T r(s) ds} S(T) \right] = E_t^T[S(T)]$$

Note that the short rate $r(s)$ might be stochastic.



Only if the short rate is deterministic, then the forward and the futures price coincide:

$$F(t, T) = E_t^T[S(T)] = E_t^{\mathbb{Q}}[S(T)] = f(t, T)$$

Convexity

If the short rate is assumed to be stochastic, then the forward and the futures prices differ

$$F(t, T) = E_t^T[S(T)] \neq E_t^Q[S(T)] = f(t, T)$$

The difference between futures price and forward price is known as *futures correction* or *convexity adjustment*.

The *convexity adjustment* is derived below

$$f(t, T) - F(t, T) = E_t^Q[S(T)] - \frac{1}{D(t, T)} E_t^Q[e^{-\int_t^T r(s) ds} S(T)] = - \frac{\text{Cov}_t^Q(e^{-\int_t^T r(s) ds}, S(T))}{D(t, T)}$$

In particular, we notice that this adjustment is zero when:

- The short rate is deterministic
- The short rate and the futures underlying are uncorrelated

If short interest rate is assumed deterministic, thus futures price coincides with the forward price of the equity at the maturity of the contract – so you could say that Futures are like “zero strike forwards” where the “payoff is not discounted”.

Why do people trade Futures

Futures are D1 contracts which offer a payoff profile and characteristics very popular amongst investors.. for obvious reasons:

1. Exposure: investor gains (quasi) D1 exposure to the underlying asset. Not much difference from a plain purchase.
2. Funding: futures trade on margin (IM/VM), only a fraction of capital is required. Cash can be deployed in alternative shells.
3. Market access: provide easy access to markets which are either hard to reach (e.g. commodities, EM indices).
4. Balance Sheet: only MtM shows on BS. Since futures are daily margined, they are almost transparent to BS.
5. Leverage: since futures trade on margin, investors can take levered positions easily.
6. Liquidity: frequently most liquid venue, allows to take large positions and facilitates hedging.
7. Counterparty risk: being exchange traded and daily margined greatly reduces counterparty risk issues, CVA and related provisions.
8. Margin management: much easier management of margin exchange vs. products subject to Non-Cleared-Derivative-Margin rules.
9. MIFID: record keeping MIFID requirements are imposed only on the exchange (MTF trading venue).
10. Physical delivery: which might be desirable. Applicable to physically settled contracts - not relevant for equity products.
11. Packaged: cum(ex) div exposure available within the product, allows to take exposure in a convenient fashion, like ETFs.
12. Can be cheap: funding can be quite cheap compared to alternative markets (e.g. synth, margin loans, repo, etc.)

Exchange Rate is Just a Price!

- The foreign exchange (FX or FOREX) market is the market where exchange rates are determined.
- Exchange rates are the mechanisms by which world currencies are tied together in the global marketplace, providing the price of one currency in terms of another.
- An exchange rate is a price, specifically the relative price of two currencies.
- For example, the U.S. Dollar/Eur exchange rate is the price of a euro expressed in U.S. dollars. On March 23, 2023, this exchange rate was USD 1.0869 per EUR, or, in market notation, 1.0869 USD/EUR.

How this affects other economic variables?

Spot FX Market

- The spot market is the exchange market for payment and delivery today.
- Currencies traded in the market have customary settlement in two business days.
- Example: JPM buys GBP 1M in the spot market at $St = 1.20$ USD/GBP. In 2 business days, JPM will receive a GBP 1M deposit and will transfer to the counterparty a USD 1.2M deposit

Quotations.

- Indirect quotation: JPY/USD (European quote).

Suppose a U.S. tourist wishes to buy JPY at Newark International Airport. A quote of JPY 110.34-111.09 means the dealer is willing to buy one USD for JPY 110.34 (bid) and sell one USD for JPY 111.09 (ask). For each USD that the dealer buys and sells, he/she makes a profit of JPY .75.

- Direct quotation: USD/JPY (American quote).

If the dealer at Newark International Airport uses direct quotations, the bid-ask quote will be USD.009002-.009063 per one JPY.

How is the quotation for non-USD ccys?

Spot FX Market (Contd.) – Cross Rates

- Lets say Singaporean trader calls a Hong Kong bank and asks for the JPY/CHF quote, the Hong Kong bank will quote a rate that does not fit under either quote system.
- Most currencies are quoted against the USD, so that cross-rates are calculated from USD quotations. For example, the EUR/GBP is calculated using the USD/EUR and USD/GBP rates.
- Suppose XYZ Bank gives the following quotes $S_t = .0104-.0108$ USD/JPY, and $S_t = 1.5670-1.5675$ USD/GBP.

XYZ Bank wants to calculate the JPY/GBP cross-rates.

The JPY/GBP bid rate is the price at which XYZ Bank is willing to buy GBP against JPY, i.e., the number of JPY units it is willing to pay for one GBP.

Buy GBP and Sell JPY is equivalent to

Sell JPY to buy 1 USD at bid rate of $(1/.0108)$ JPY/USD

Resell USD to buy 1 GBP at bid rate of 1.5670 USD/GBP

$$S_{bid,JPY/GBP} = S_{bid,JPY/USD} * S_{bid,USD/GBP}$$

$$[(1/.0108) \text{ JPY/USD}] * [(1.5670) \text{ USD/GBP}] = 145.0926$$

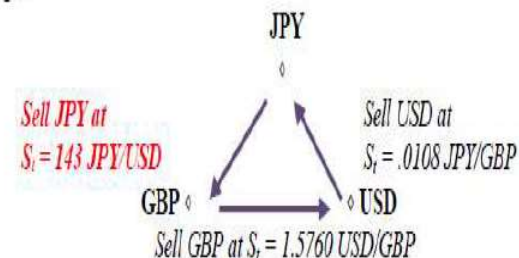
$$S_{ask,JPY/GBP} = S_{ask,JPY/USD} * S_{ask,USD/GBP}$$

$$[(1/.0104) \text{ JPY/USD}] * [(1.5675) \text{ USD/GBP}] = 150.7211$$

What kind of Arbitrage can occur here?

Lets say we observe quote of 143 JPY/USD -> Triangle Arb!!

The triangle:



This operation makes a profit of USD $14,633.51$ (or 1.46% per USD borrowed).

Forward FX Market

- Spot FX but with differed settlement!! Fix the currency price today for future selling of the same

- Example

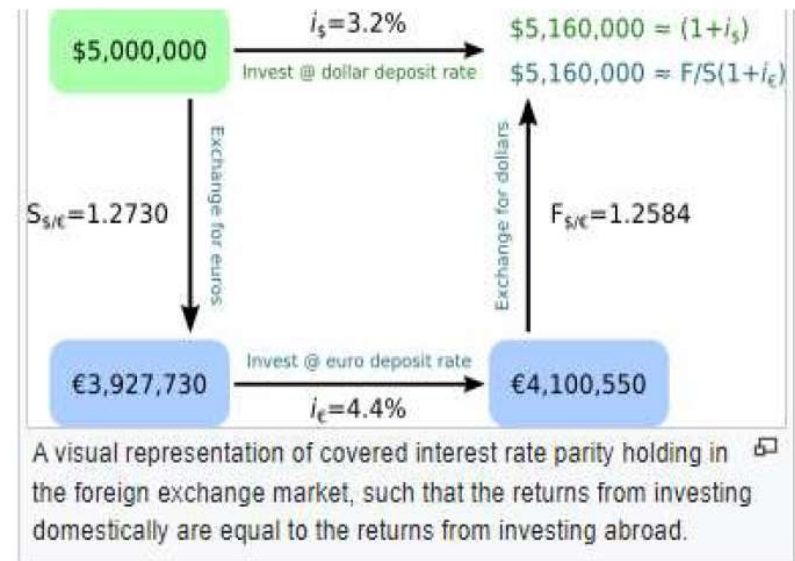
BOFA sells GBP 1M forward using a 7-day GBP forward contract at $F_{t,7\text{-day}} = 1.205$ USD/GBP. In 7 days, BOFA will receive a USD 1,205,000 deposit and will transfer to the counterparty a GBP 1M deposit.

- Valuation (using Covered Interest Rate Parity)

“Forward exchange rate sustains equilibrium such that the dollar return on dollar deposits is equal to the dollar return on foreign deposit, thereby eliminating the potential for covered interest arbitrage profits”

What is Uncovered IRP?

Is Covered IRP always hold true?



$$(1 + i_d) = \frac{F}{S} * (1 + i_f)$$

where:

i_d = The interest rate in the domestic currency or the base currency

i_f = The interest rate in the foreign currency or the quoted currency

S = The current spot exchange rate

F = The forward foreign exchange rate

QUESTIONS!!